ALA 1

Cloud Task Manager on AWS

Deploying a web application with compute and storage services

Student	Vanshvi Patel
Activity	CNS ALA 1: cloud deployment of a basic web application
Cloud platform	Amazon Web Services (AWS)
Region	Asia Pacific (Sydney), ap-southeast-2
Live URL	http://13.237.151.164
Project page	http://13.237.151.164/project.html
Source code	https://github.com/Vanshvi06/cns-ala1-cloud
Deployed on	1 November 2026

1. Aim

To create and deploy a basic web application on a cloud platform, configure a compute service and a storage service, make the application reachable over the internet, and demonstrate basic resource management.

2. Deployment summary

A task manager web application runs on an Amazon EC2 virtual machine. Uploaded files are stored in Amazon S3 and tasks are stored on the instance's EBS disk. The application is reachable through a fixed Elastic IP address.

Compute service	Amazon EC2 instance ala1-server (i-09312cbb21eb43e7e), t3.micro, Ubuntu Server 24.04 LTS
Web server	Nginx reverse proxy on port 80, forwarding to a Node.js (Express) app on port 3000, kept running by PM2
Storage service	Amazon S3 bucket ala1-storage-vanshvi-2026 (uploaded files) and an 8 GiB gp3 EBS volume (tasks.json)
Access control	IAM role ala1-ec2-s3-role with AmazonS3FullAccess; security group ala1-sg allowing SSH (22) and HTTP (80)
Public address	Elastic IP 13.237.151.164 attached to the instance
Resource management	CloudWatch CPU graph, AWS Budget ala1-budget ($1.00), Elastic IP, stop and start test, live statistics inside the app

Note: some screenshots were taken before the Elastic IP was attached and show the earlier public IP 15.134.186.197. The final address of the application is 13.237.151.164.

3. Application and architecture

The application has four parts on one page: a task manager (add, complete and delete tasks), a file upload area connected to Amazon S3, a Resource Management section showing live CPU, memory and disk usage from the server, and a Deployment Details section.

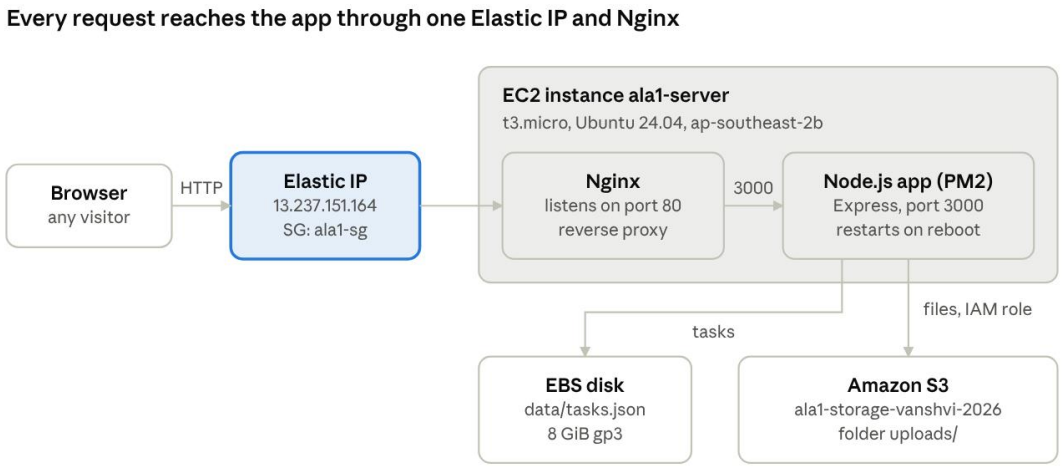


Figure 1: Request flow: browser, Elastic IP, Nginx, Node.js app, EBS disk and Amazon S3

4. Source code on GitHub

The code is kept in the public repository cns-ala1-cloud and was pushed from Git Bash to the main branch.

Repository: https://github.com/Vanshvi06/cns-ala1-cloud

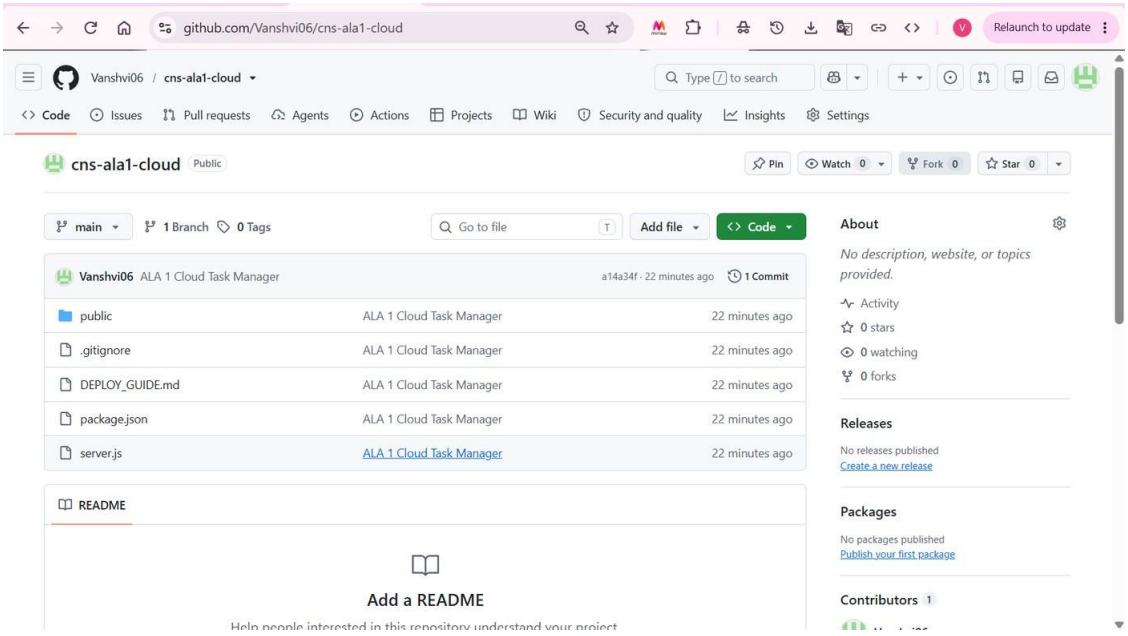


Figure 2: GitHub repository cns-ala1-cloud with the application files

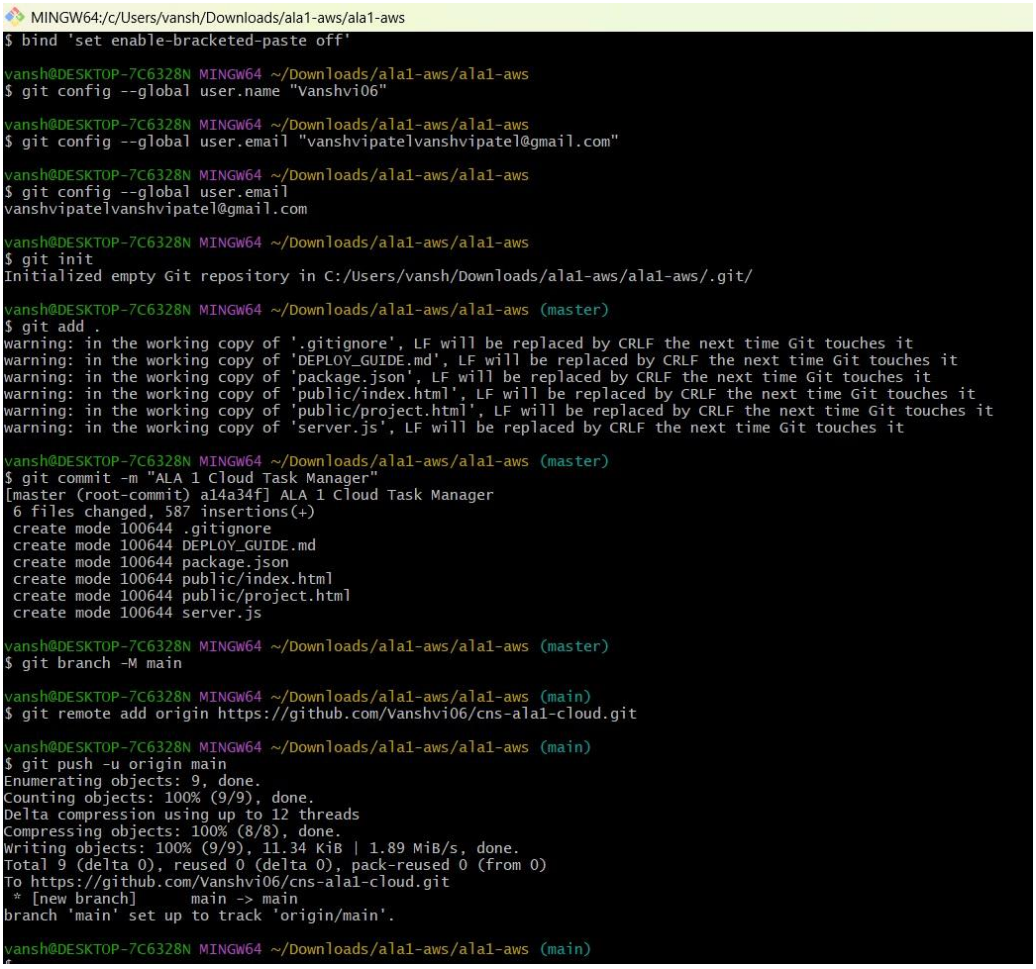


Figure 3: Code pushed to GitHub from Git Bash

5. Compute service: Amazon EC2

The EC2 instance was launched in the Sydney region and configured through the browser-based EC2 Instance Connect terminal.

1. Launched the instance ala1-server (t3.micro, Ubuntu Server 24.04 LTS, 8 GiB gp3) with key pair ala1-key and security group ala1-sg.
2. Attached the IAM role ala1-ec2-s3-role so the app can reach S3 without access keys.
3. Connected with EC2 Instance Connect and updated the system packages.
4. Installed Node.js 20, Nginx and Git, cloned the repository and ran npm install.
5. Started the app with PM2, setting S3_BUCKET and AWS_REGION, and enabled start on boot with pm2 startup and pm2 save.
6. Configured Nginx to forward port 80 to the app on port 3000 and checked the configuration with nginx -t.

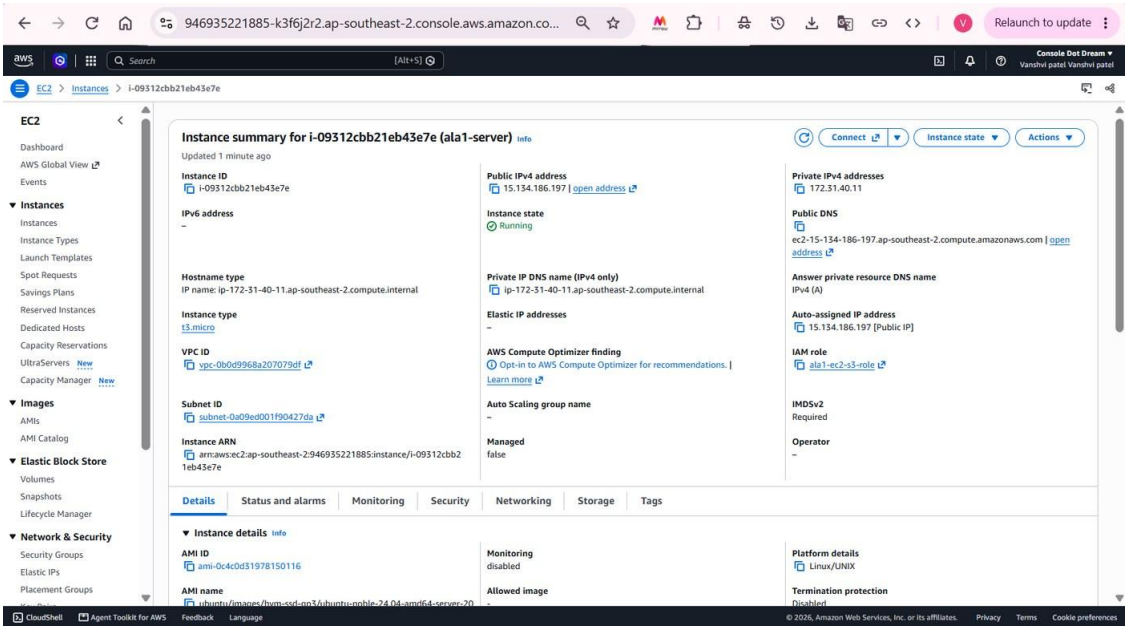


Figure 4: EC2 instance ala1-server running, with its public IP and IAM role

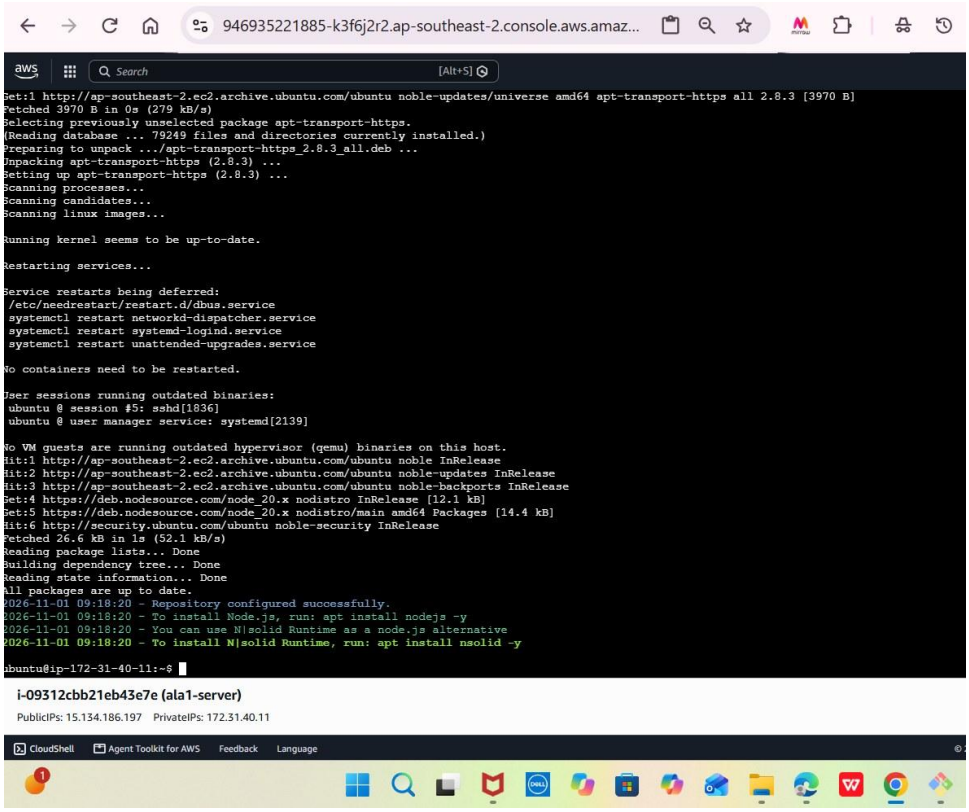


Figure 5: Terminal session on the instance through EC2 Instance Connect

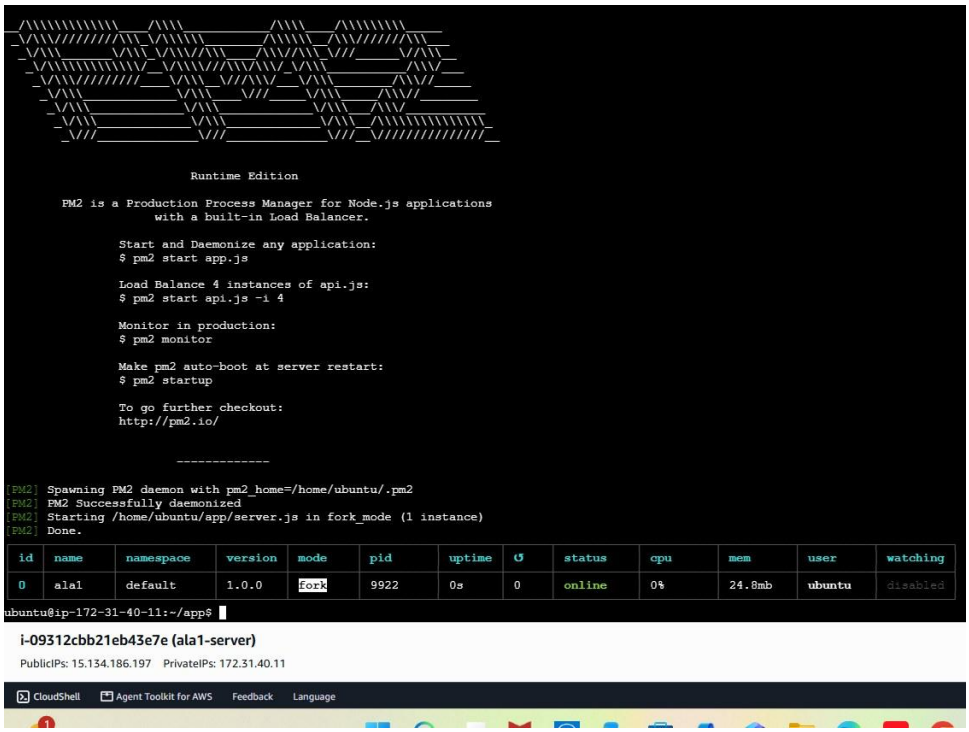


Figure 6: Application process ala1 online under PM2

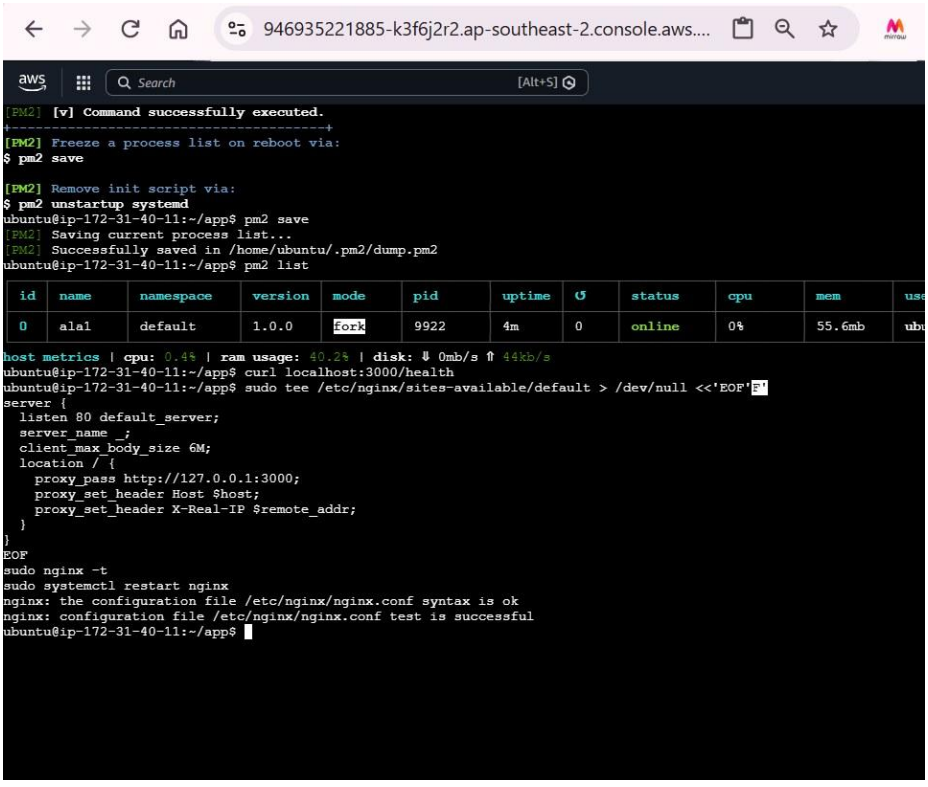


Figure 7: PM2 process list and a successful Nginx configuration test

6. Storage service: Amazon S3

Files uploaded in the app are saved in an Amazon S3 bucket created in ap-southeast-2 with Block all public access turned on. The app stores files in the uploads/ folder and lists them through signed links that expire after 5 minutes. The IAM role ala1-ec2-s3-role, with the AWS managed policy AmazonS3FullAccess, is attached to the instance.

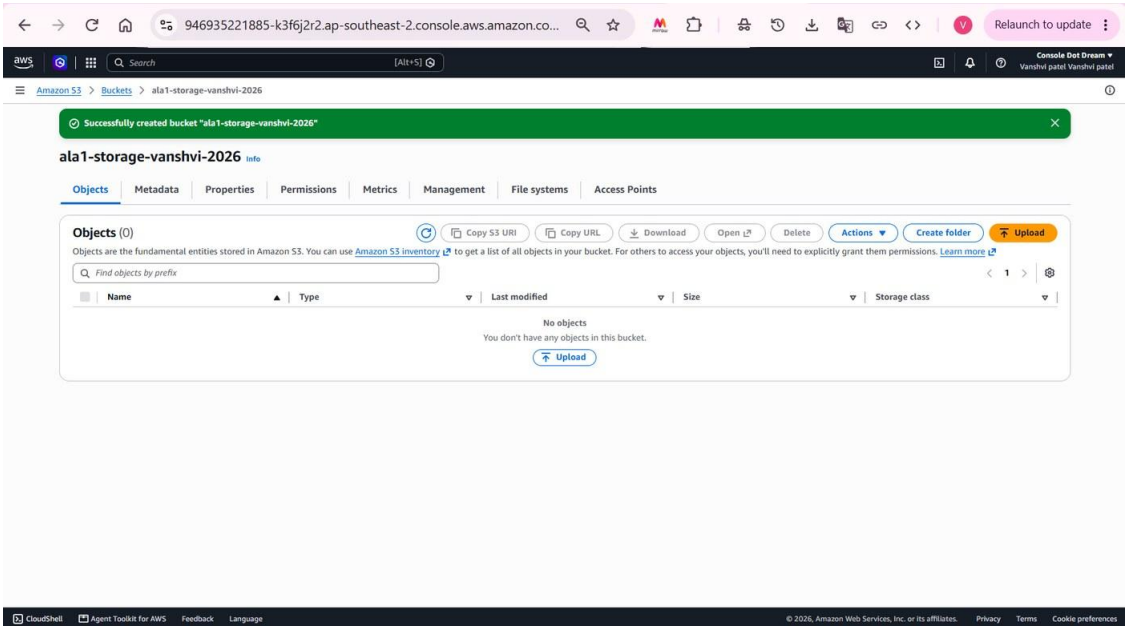


Figure 8: S3 bucket ala1-storage-vanshvi-2026 created

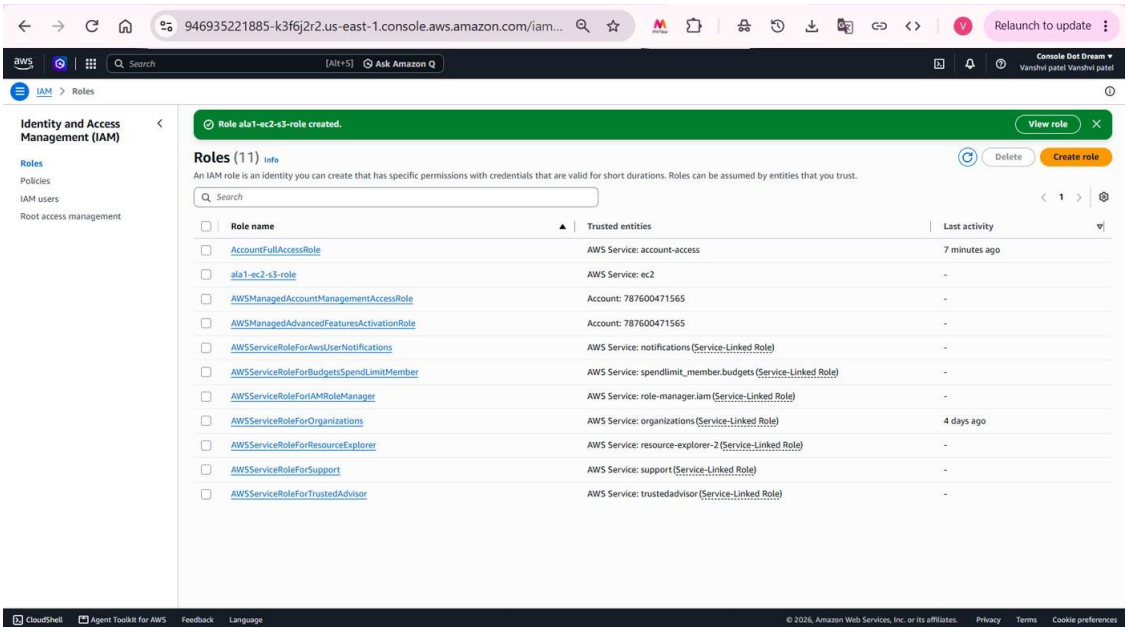


Figure 9: IAM role ala1-ec2-s3-role created

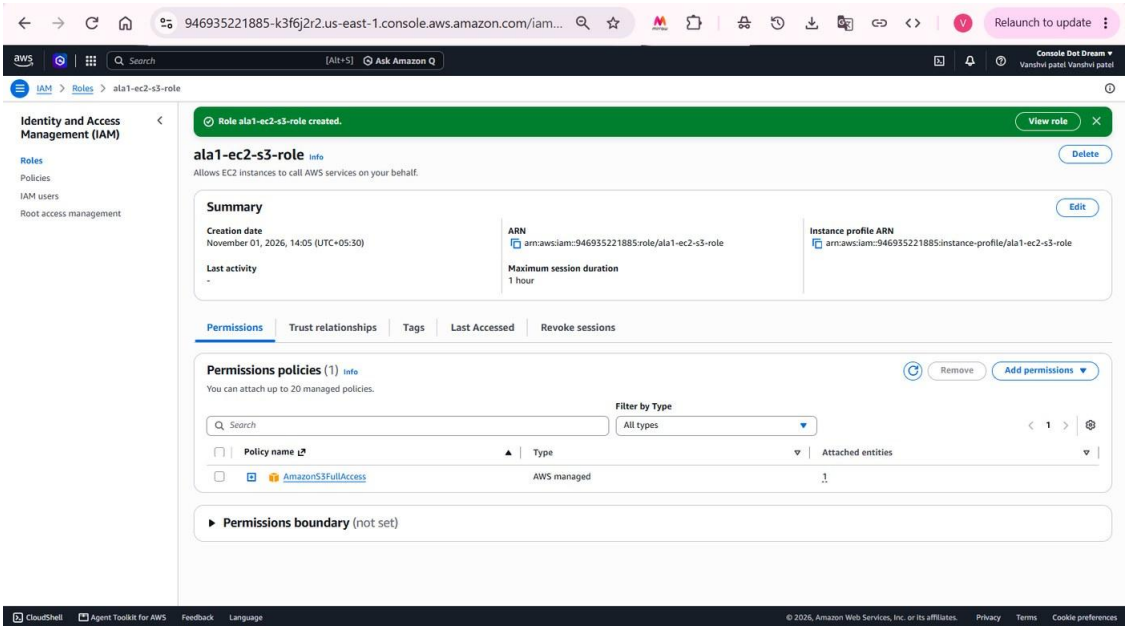


Figure 10: AmazonS3FullAccess attached to the role

Upload test: the file nature.jpg (94 KB) was uploaded in the app and then found in the uploads/ folder of the bucket.

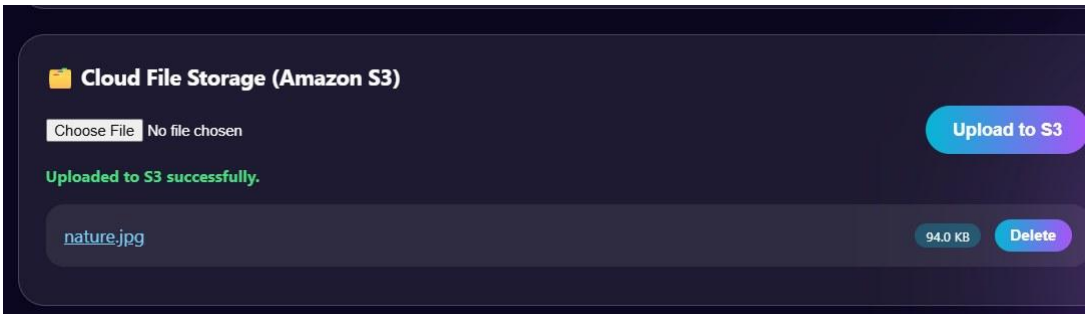


Figure 11: File uploaded to S3 and listed in the app

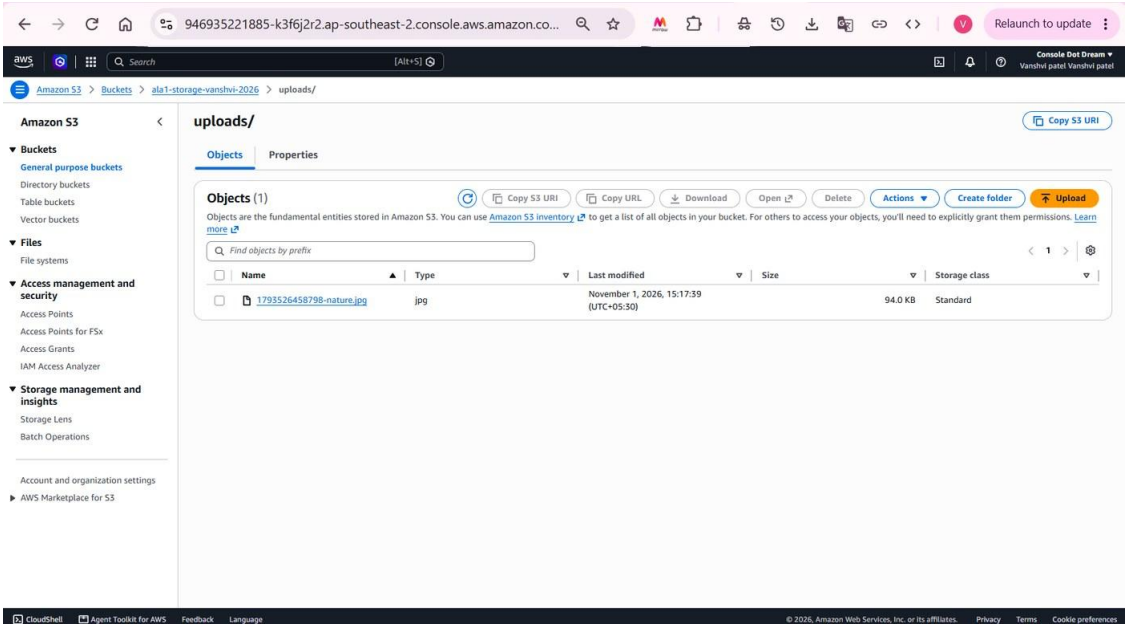


Figure 12: The uploaded file inside the S3 bucket

7. Accessibility over the internet

The security group ala1-sg allows HTTP on port 80 and SSH on port 22 from any address. A browser request reaches Nginx through the Elastic IP, and Nginx passes it to the Node.js app. The site uses HTTP only; HTTPS is not configured.

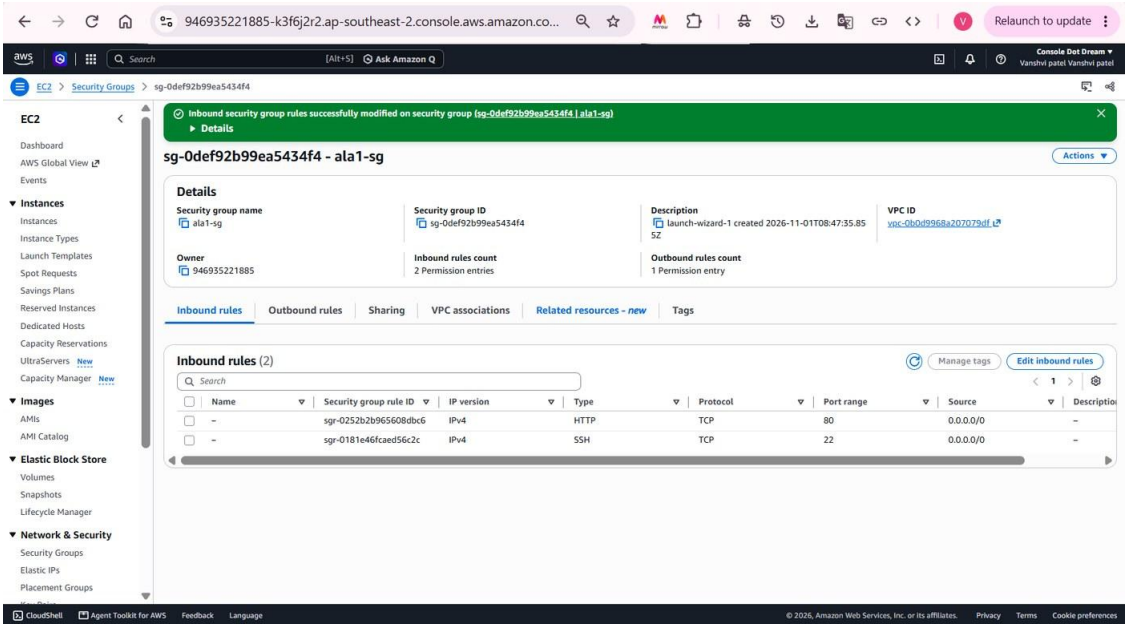


Figure 13: Inbound rules of security group ala1-sg (ports 22 and 80)

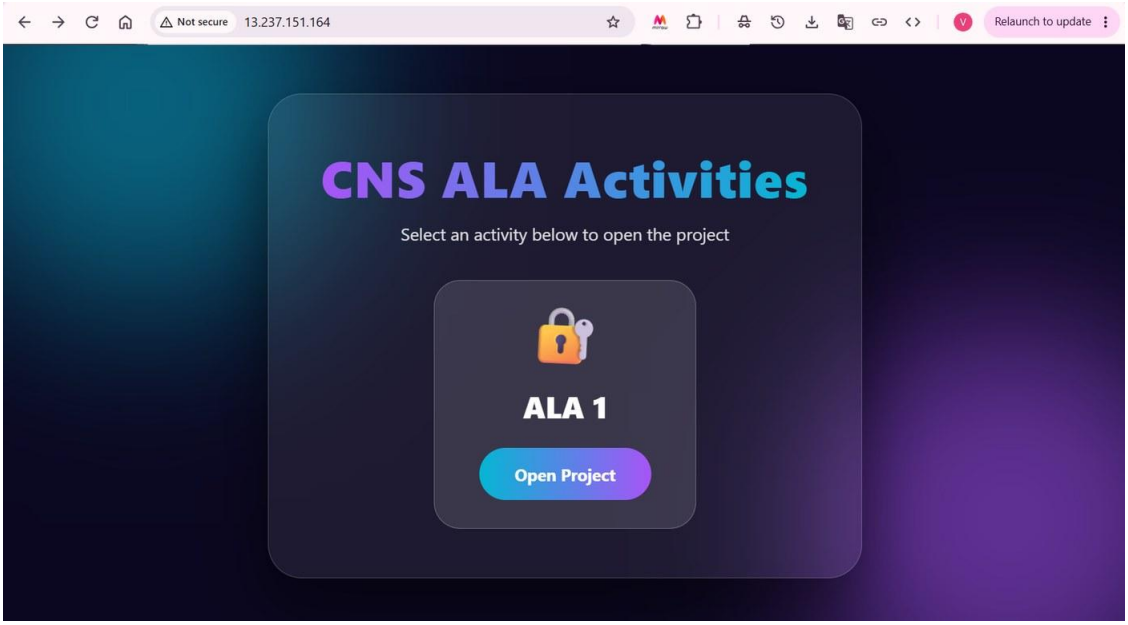


Figure 14: Home page opened in a browser at http://13.237.151.164

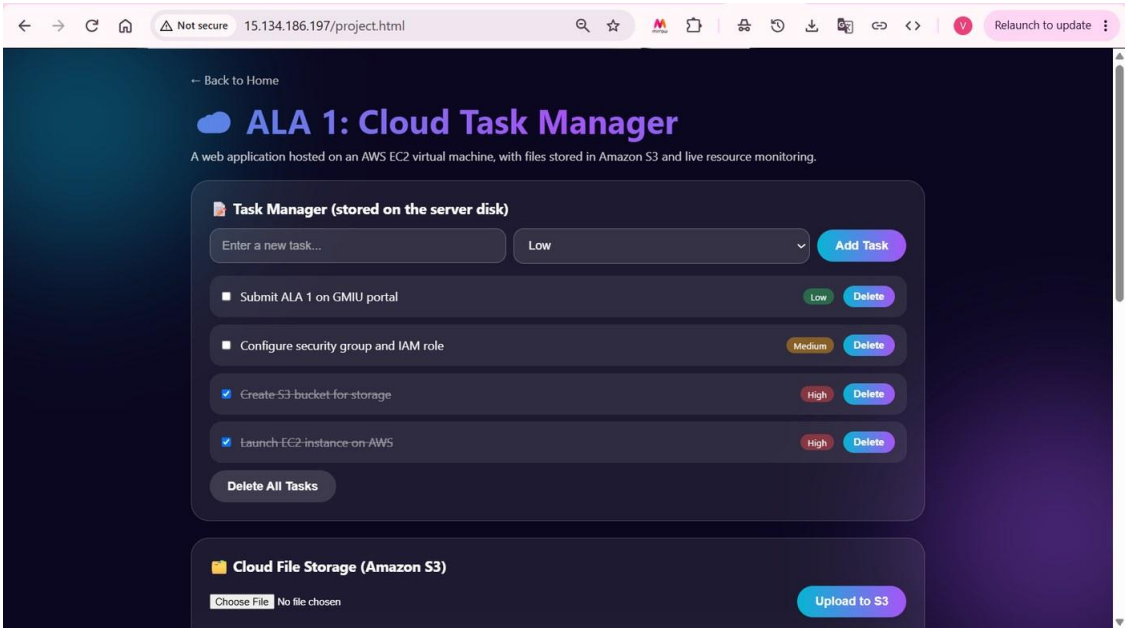


Figure 15: Project page with tasks added and completed, and the upload section

8. Resource management

- Server resources: disk and memory usage checked on the server with df -h and free -h.
- Monitoring: the CloudWatch CPU utilization graph for the instance, plus live CPU, memory and disk statistics inside the app.
- Cost control: AWS Budget ala1-budget with a limit of $1.00.
- Fixed address: Elastic IP 13.237.151.164 attached to the instance.
- Restart test: the instance was stopped and started; it returned to Running with 3/3 status checks passed, kept the same IP, and the app, tasks and files were available again.

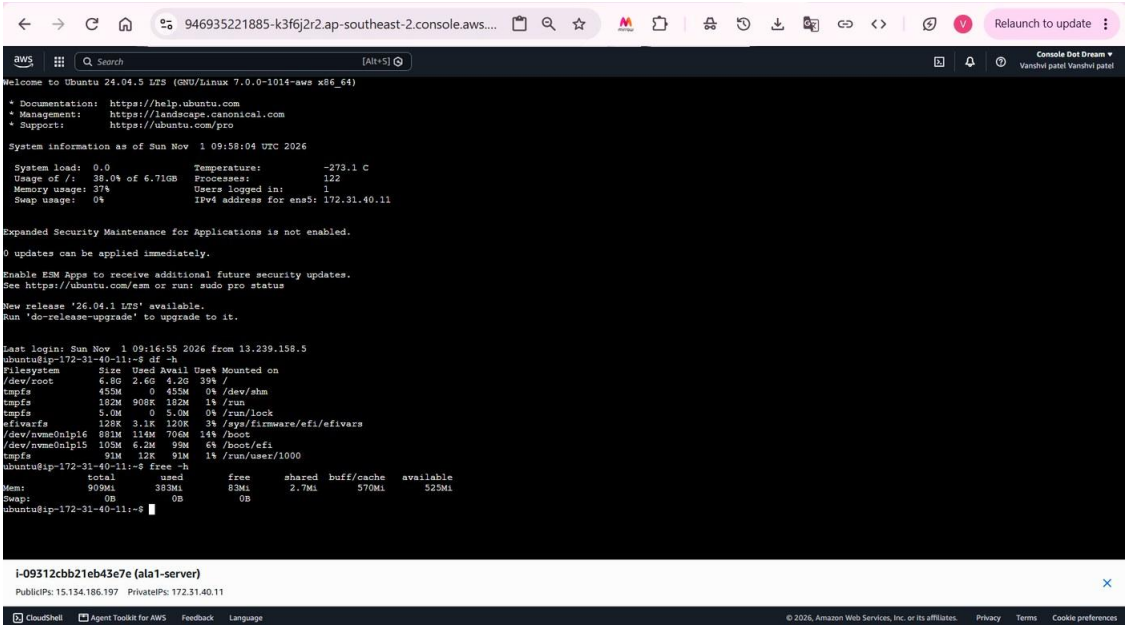


Figure 16: Disk and memory usage on the server

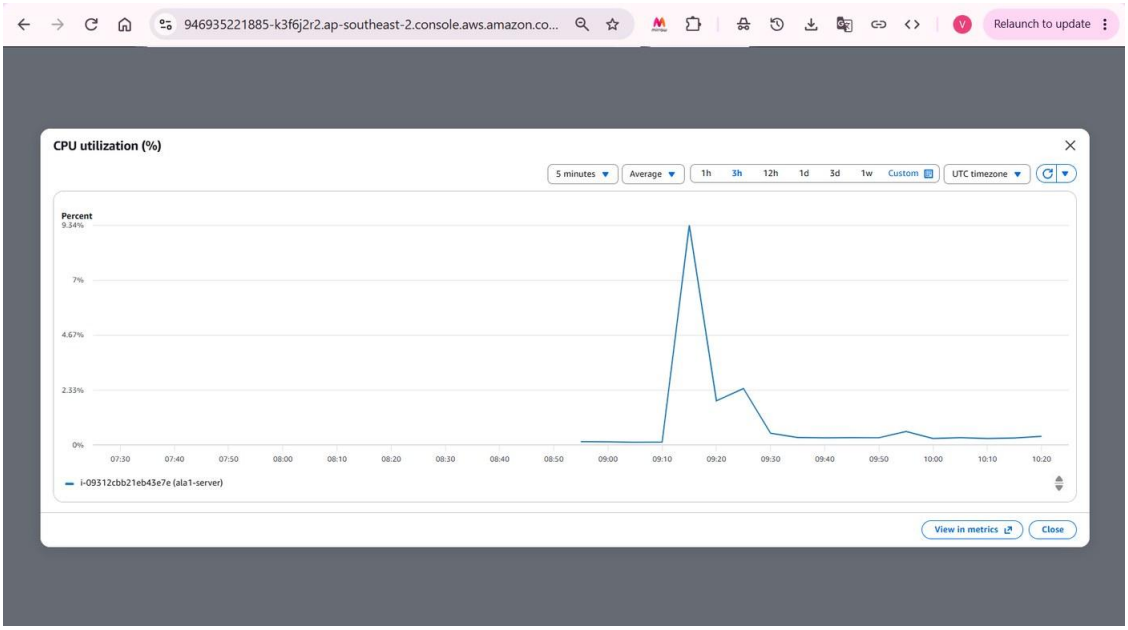


Figure 17: CloudWatch CPU utilization of the instance

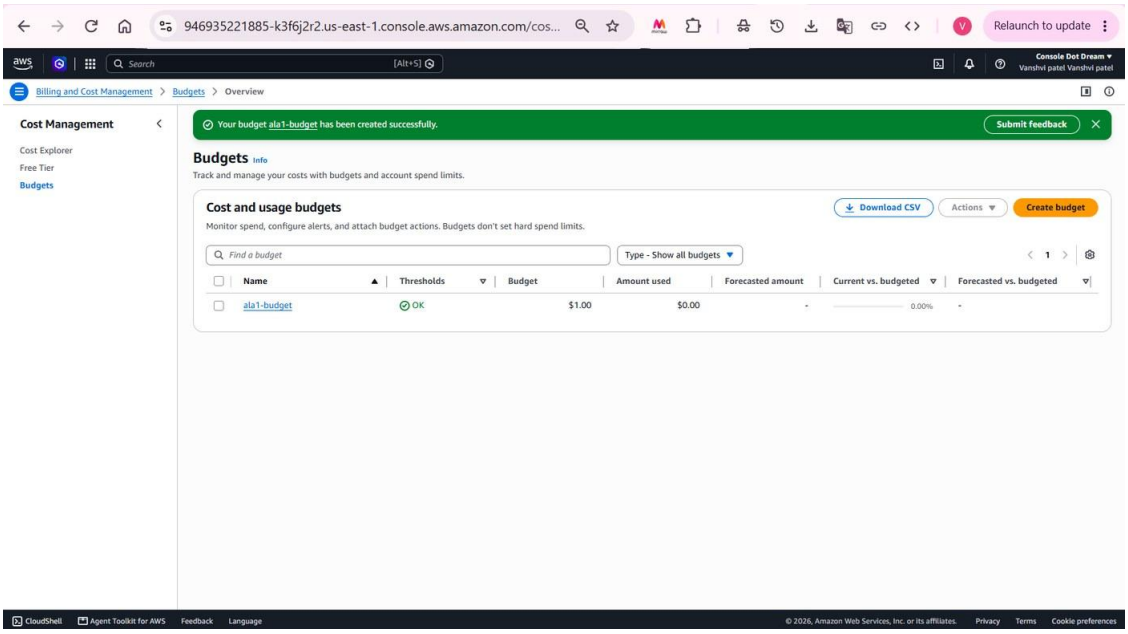


Figure 18: AWS Budget ala1-budget created

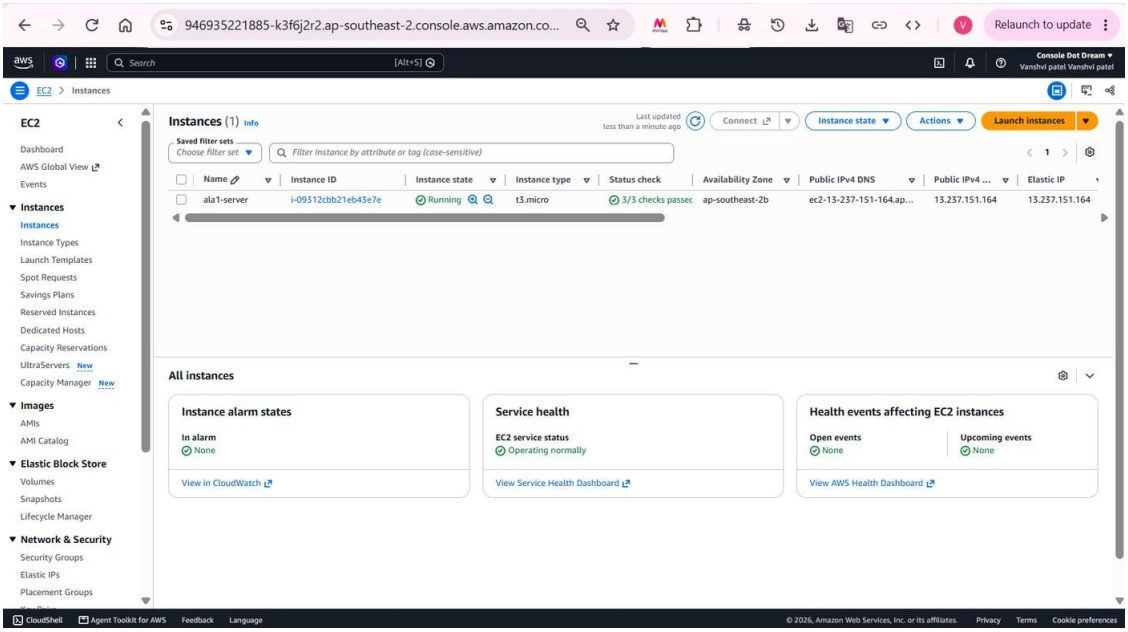


Figure 19: Instance running with 3/3 checks passed and Elastic IP 13.237.151.164 after the restart

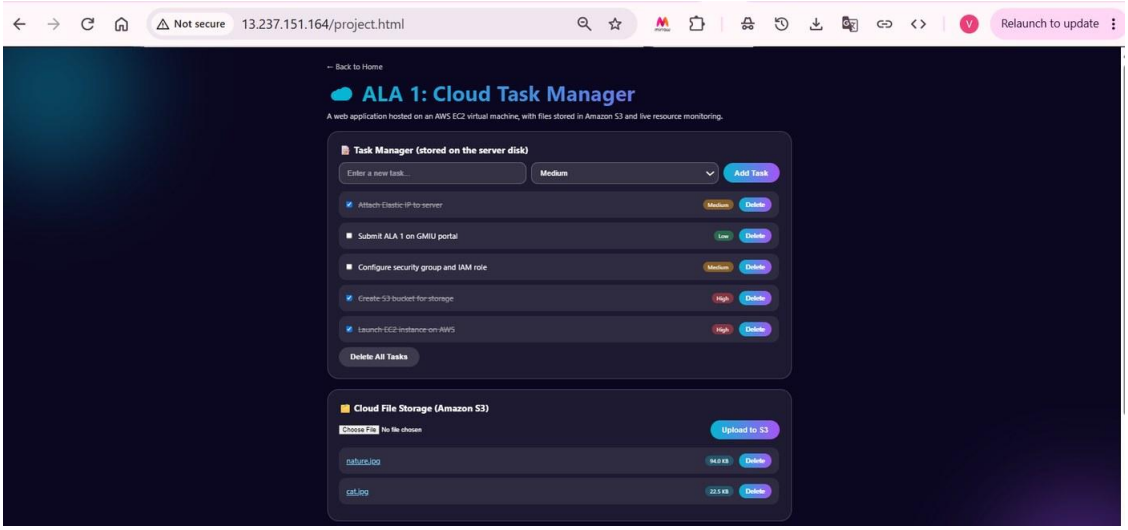


Figure 20: Tasks and files still available after the restart

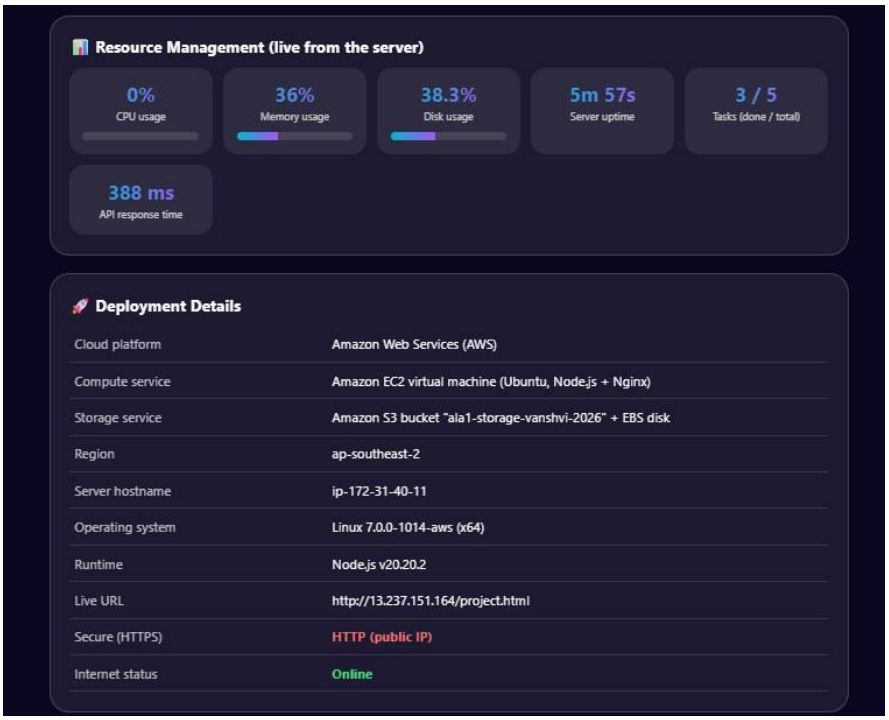


Figure 21: Resource Management and Deployment Details sections of the app

9. Conclusion

The task manager was deployed on an Amazon EC2 instance, stores uploaded files in Amazon S3 and tasks on the EBS disk, and is reachable over the internet at http://13.237.151.164. Resource use is watched through CloudWatch, a budget alert and the live statistics in the app, and the application recovered on its own after a stop and start of the instance.